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Past Papers by Waqar Siddhu
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A sub graph of a graph G that contains every vertex of G and is a tree is called

VuAnswers.com

Answer (Please select your correct option)

Trivial tree

☐

empty tree

☐

Spanning tree

☐

correct

Made by: Waqar Siddhu

A vertex of degree greater than 1 in a tree is called a

VuAnswers.com

Answer (Please select your correct option)

Branch vertex

☐

correct

Terminal vertex

☐

Ancestor

☐

Made by: Waqar Siddhu

A circuit that consist of a single vertex is called

VuAnswers.com

Answer (Please select your correct option)

correct

Trivial



Tree



Empty



Made by: Waqar Siddhu

If a graph is a tree then

VuAnswers.com

Answer (Please select your correct option)

☐ it has 2 spanning trees

☒ it has only 1 spanning tree

☐ it has 4 spanning trees

☐ it has 5 spanning trees

correct

Made by: Waqar Siddhu

The logical expression $p \vee q$ will be read as

VuAnswers.com

Answer (Please select your correct option)

correct

☐ $p \text{ or } q$

☐ $p \text{ and } q$

☐ $p \times q$

☐ $p - q$

Made by: Waqar Siddhu

If p = It is red,
 q = It is hot
Then "It is not red but hot" is denoted by : $p \wedge : q$.

Answer (Please select your correct option)

VuAnswers.com

☐ True

correct

☐ False

Made by: Waqar Siddhu

If A, B and C are the subsets of a universal set U then $(A \cup B) \cup C$ is equal to

VuAnswers.com

Answer (Please select your correct option)

☐

$$A \cup (B \cap C)$$

☐

$$A \cap (B \cup C)$$

☐

$$\emptyset$$

☐

$$A \cup (B \cup C)$$

correct

Made by: Waqar Siddhu

The power set of a set A is the set of all subsets of A and its denoted by $P(A)$.

Answer (Please select your correct option)

VuAnswers.com

False



True



correct

Made by: Waqar Siddhu

Identify the Associative law of union for three sets

VuAnswers.com

Answer (Please select your correct option)

☐

$$A \cup (B \cup C) = (A \cup B) \cup C$$

correct

☐

$$A \cap (B \cap C) = (A \cap B) \cap C$$

☐

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

☐

None of these

Made by: Waqar Siddhu

Let f and g be the functions defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$ then composition of f and g is

VuAnswers.com

Answer (Please select your correct option)

☐ $6x + 6$

☐ $5x + 5$

☒ $6x + 7$

correct

Made by: Waqar Siddhu

If $f(x) = 2x + 1$, $g(x) = x^2 - 1$ then $f \circ g(x) =$

VuAnswers.com

Answer (Please select your correct option)

☐

$x^2 - 1$

☐

$2x^2 - 1$

correct

☐

$2x^3 - 1$

Made by: Waqar Siddhu

Let f is defined recursively by $f(0) = 3, f(n+1) = 2f(n) + 3$ then $f(1) =$

VuAnswers.com

Answer (Please select your correct option)

☐ 9 correct

☐ 10

☐ 18

☐ 21

Made by: Waqar Siddhu

If $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ for all integers $n \geq 1$ then $P(k)$ is

VuAnswers.com

Answer (Please select your correct option)

☐ $1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$

☐ $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

☐ $1 + 2 + 3 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}$

correct

☐ $1 + 2 + 3 + \dots + (k-1) = \frac{k(k-1)}{2}$

Made by: Waqar Siddhu

In Mathematical Induction, inductive step is

VuAnswers.com

Answer (Please select your correct option)

☐ $\forall k, P(k) \rightarrow P(n)$

☐ $\forall k, P(k) \rightarrow P(k+1)$

correct

☐ $\forall k, P(k) \rightarrow P(n+1)$

☐ $\forall k, P(k) \rightarrow P(k-1)$

Made by: Waqar Siddhu

$P(n)$ is called statement or

VuAnswers.com

Answer (Please select your correct option)

☐ sentence

☐ proposition

☐ inequality

☐ none of these

correct

Made by: Waqar Siddhu

For all positive integer values of n , $5^n - 1$ is divisible by

VuAnswers.com

Answer (Please select your correct option)

☐

3

☐

4

correct

☐

6

☐

0

Made by: Waqar Siddhu

An integer n is prime if and only if $n > 1$ and for all positive integers r and s , if $n = r.s$ then

VuAnswers.com

Answer (Please select your correct option)

☐ $r = 1$ and $s = 2$

correct

☐ $r = 1$ and $s = 0$

☐ $r = 2$ and $s = 3$

☐ None of these

Made by: Waqar Siddhu

Quotient Remainder Theorem states that for any positive integer d , there exist unique integer q and r such that ----- and $0 \leq r < d$.

Answer (Please select your correct option)

VuAnswers.com

☐

$$n = d.r + q$$

☐

$$n = d.q + r$$

correct

☐

$$n = r.q + d$$

☐

None of these

Made by: Waqar Siddhu

The word ----- refers to a step-by-step method for performing some action.

VuAnswers.com

Answer (Please select your correct option)

☐ Series

☐ Relation

☐ Algorithm

☐ Function

correct

Made by: Waqar Siddhu

The greatest common divisor of 27 and 72 is

VuAnswers.com

Answer (Please select your correct option)

☐ 27

☒ 9

☐ 1

☐ None of these

correct

Made by: Waqar Siddhu

In how many ways can a set of five letters be selected from the English Alphabets?

VuAnswers.com

Answer (Please select your correct option)

☐ $C(26, 5)$

correct

☐ $C(5, 26)$

☐ $C(12, 3)$

☐ None of these

Made by: Waqar Siddhu

If one event can occur in n_1 ways and if for each of these n_1 ways, a second event can occur in n_2 ways, then the total number of ways in which both events occur is

VuAnswers.com

Answer (Please select your correct option)

☐ $n_1 + n_2$

☐ $n_1 \cdot n_2$

correct

☐ $P(n)_1 \cdot P(n)_2$

☐ $P(n_1) + P(n_2)$

Made by: Waqar Siddhu

An arrangement of objects without the consideration of order is called

VuAnswers.com

Answer (Please select your correct option)

☐ Permutation

☒ Combination

☐ Selection

☐ None of these

correct

Made by: Waqar Siddhu

If the order does not matter and repetition is allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)



n^k



$C(n+k-1, k)$

correct



$P(n, k)$



$C(n, k)$

Made by: Waqar Siddhu

If the order matters and repetition is not allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)

☐

$$n^k$$

☐

$$C(n+k-1, k)$$

☐

$$P(n, k)$$

correct

☐

$$C(n, k)$$

Made by: Waqar Siddhu

$A = \{1, 2, 3, 4, 5\}$ is a set of first five ----- numbers.

VuAnswers.com

Answer (Please select your correct option)

☐ True

☐ natural

correct

☐ even

☐ odd

Made by: Waqar Siddhu

Among 200 people, 150 either play tennis or snooker or both. If 85 play tennis and 60 play tennis and snooker, how many play snooker?

Answer (Please select your correct option)

VuAnswers.com

☐ 125

☐ 225

☐ 85

☐ 25

correct

Made by: Waqar Siddhu

A procedure that yields a given set of possible outcomes is called

VuAnswers.com

Answer (Please select your correct option)

Event

☐

Outcome

☐

Experiment

☐

correct

Made by: Waqar Siddhu

What is the probability of getting a number greater than 4 when a die is thrown?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{2}$

☐ $\frac{3}{2}$

☐ $\frac{1}{3}$

☐ 1

correct

Made by: Waqar Siddhu

If two fair dice are thrown, what is the probability of getting a double six?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{6}$

☒ $\frac{1}{36}$

correct

☐ 1

☐ $\frac{1}{2}$

Made by: Waqar Siddhu

If A and B are two disjoint (mutually exclusive) events then $P(A \cup B) =$

VuAnswers.com

Answer (Please select your correct option)

☐

$$P(A) + P(B) + P(A \cap B)$$

☐

$$P(A) + P(B) + P(A \cup B)$$

☐

$$P(A) + P(B) - P(A \cap B)$$

☐

$$P(A) + P(B)$$

correct

Made by: Waqar Siddhu

If a fair die is thrown then what is the probability that a prime number appear?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{2}$

correct

☐ $\frac{1}{6}$

☐ $\frac{1}{3}$

☐ $\frac{2}{3}$

Made by: Waqar Siddhu

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

VuAnswers.com

Answer (Please select your correct option)

Dependent

☐

correct

Independent

☐

Exclusive

☐

Made by: Waqar Siddhu

If A, B and C are any three events, then $P(A \cup B \cup C)$ is equal to

VuAnswers.com

Answer (Please select your correct option)

☐

$$P(A) + P(B) + P(C)$$

☐

$$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

correct

☐

$$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$$

☐

$$P(A) + P(B) + P(C) + P(A \cap B \cap C)$$

Made by: Waqar Siddhu

How many vertices will the graph have if it contain 16 edges and all vertices of degree 2?

VuAnswers.com

Answer (Please select your correct option)

☐ 14

☒ 16

☐ 18

☐ 32

correct

Made by: Waqar Siddhu

If a graph has any vertex of degree 3 then

VuAnswers.com

Answer (Please select your correct option)

It must have Euler circuit

☐

It must have Hamiltonian circuit

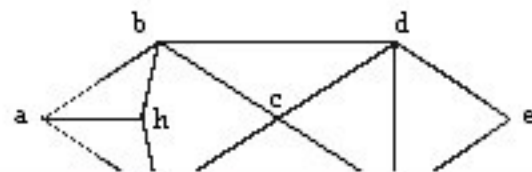
☐

It does not have Euler circuit

☐

Made by: Waqar Siddhu

The Hamiltonian circuit for the following graph is



Answer (Please select your correct option)

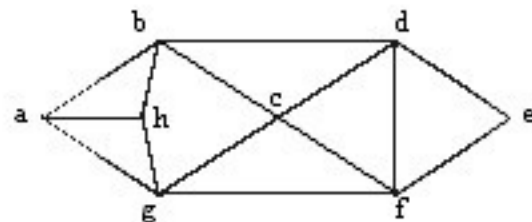
VuAnswers.com

☐ abcdefgh

☐ abefgha

☐ abcdefgha

Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

☐ abcdefgh☐ abefgha☐ abcdefgha

Made by: Waqar Siddhu

Changing rows of a matrix into its columns is called

VuAnswers.com

Answer (Please select your correct option)

☐ symmetric matrix

☐ transpose of matrix

☐ adjoint of matrix

☐ Hermitian Matrix

Made by: Waqar Siddhu

Two matrices are said to be conformable for multiplication if number of(a)..... of 1st matrix is equal to number of(b)..... in 2nd matrix

Answer (Please select your correct option)

VuAnswers.com

☐ (a) rows, (b) columns

☐ (a) columns, (b) rows

☐ (a) columns, (b) columns

☐ (a) rows, (b) rows

Made by: Waqar Siddhu

A sub graph of a graph G that contains every vertex of G and is a tree is called

VuAnswers.com

Answer (Please select your correct option)

☐ Trivial tree

☐ empty tree

☒ Spanning tree

correct

Made by: Waqar Siddhu

A vertex of degree 1 in a tree is called a

VuAnswers.com

Answer (Please select your correct option)

☐ Terminal vertex

correct

☐ Internal vertex

Made by: Waqar Siddhu

If a tree has 8 vertices then it has

VuAnswers.com

Answer (Please select your correct option)

6 edges



7 edges



correct

9 edges



Made by: Waqar Siddhu

Complete graph is planar if

VuAnswers.com

Answer (Please select your correct option)

☐ $n = 4$

☐ $n > 4$

☐ $n \leq 4$

correct

Made by: Waqar Siddhu

Any two spanning trees for a graph

VuAnswers.com

Answer (Please select your correct option)



Does not contain same number of edges



Have the same degree of corresponding edges



contain same number of edges

correct



May or may not contain same number of edges

Made by: Waqar Siddhu

If p and q are statement variables then bi-conditional of p and q is denoted by

VuAnswers.com

Answer (Please select your correct option)

☐ $p \leftrightarrow q$

correct

☐ $p \rightarrow q$

☐ $q \rightarrow p$

☐ None of these

Made by: Waqar Siddhu

Rephrase the following statement in bi-conditional form
“If you get up early in the morning, you will be healthy”

VuAnswers.com

Answer (Please select your correct option)

☐ You will be healthy if and only if you get up early in the morning

correct

☐ If you will be healthy then you will get up early in the morning

☐ If you will get up early in the morning then you will be healthy

☐ None of these

Made by: Waqar Siddhu

Which of the following law is used to show?

$$p \leftrightarrow q \equiv q \leftrightarrow p$$

Answer (Please select your correct option)

VuAnswers.com



Implication Law



Commutative law

correct



Exportation Law



None of these

Made by: Waqar Siddhu

If A and B are two sets then the set which contains all those elements that belong to A or B is

VuAnswers.com

Answer (Please select your correct option)



$A \cup B$

correct



$A \cap B$



$A - B$



None of these

Made by: Waqar Siddhu

Let g be the function defined by $g(x) = 3x + 2$ then $g \circ g(x) =$

VuAnswers.com

Answer (Please select your correct option)

☐ $9x^2 + 4$

☐ $6x + 4$

☐ $9x + 8$

correct

Made by: Waqar Siddhu

If A is a set of all integers and $R = \{(x, y) \in A \mid xy \geq 1\}$ is a relation on A then the relation R is

VuAnswers.com

Answer (Please select your correct option)

☐ Transitive

☐ Symmetric

☐ Reflexive

☐ All the given options are true

correct

Made by: Waqar Siddhu

The part of definition which can be expressed in terms of smaller versions of itself is called

VuAnswers.com

Answer (Please select your correct option)



Base



Restriction



Recursion



Conclusion

correct

Made by: Waqar Siddhu

Let f is defined recursively by $f(0) = 3, f(n+1) = 2f(n) + 3$ then $f(1) =$

VuAnswers.com

Answer (Please select your correct option)



9

correct



10



18



21

Made by: Waqar Siddhu

Proof of a statement by induction comprises of two basic steps:

VuAnswers.com

Answer (Please select your correct option)



Inductive and Deductive



Basis and Inductive

correct



Arranging and Sorting



None of these

Made by: Waqar Siddhu

In Mathematical Induction, inductive step is

VuAnswers.com

Answer (Please select your correct option)

☐

$$\forall k, P(k) \rightarrow P(n)$$

☒

$$\forall k, P(k) \rightarrow P(k+1)$$

correct

☐

$$\forall k, P(k) \rightarrow P(n+1)$$

☐

$$\forall k, P(k) \rightarrow P(k-1)$$

Made by: Waqar Siddhu

The indirect proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)

- ☐ Considering : q and then try to reach p
- ☒ Considering p and : q are true and try to reach contradiction
- ☐ Considering p and then try to reach q
- ☐ Considering : p and then try to reach q

correct

Made by: Waqar Siddhu

An integer n is called a perfect square for some integer k iff

VuAnswers.com

Answer (Please select your correct option)



$$n = k$$



$$n^2 = k$$



$$n = k^2$$

correct



$$n = 1$$

Made by: Waqar Siddhu

An integer n is called a perfect square for some integer k iff

VuAnswers.com

Answer (Please select your correct option)



$$n = k$$



$$n^2 = k$$



$$n = k^2$$

correct



$$n = 1$$

Made by: Waqar Siddhu

An integer n is called a perfect square for some integer k iff

VuAnswers.com

Answer (Please select your correct option)



$$n = k$$



$$n^2 = k$$



$$n = k^2$$



$$n = 1$$

Made by: Waqar Siddhu

An integer n is odd for some integer k iff

VuAnswers.com

Answer (Please select your correct option)



$$n = 2k$$



$$n = 2(k+1)$$



$$n = 2(k-1)$$



$$n = 2k+1$$

correct

Made by: Waqar Siddhu

What is the contra positive of the given statement:

If square root of every prime number is irrational then square root of 2 is irrational.

VuAnswers.com

Answer (Please select your correct option)



If square root of 2 is not irrational then square root of every prime number is not irrational.

correct



If square root of 2 is irrational then square root of every prime number is not irrational.



If square root of 2 is not irrational then square root of every prime number is irrational.



If square root of every prime number is not irrational then square root of 2 is not irrational.

Made by: Waqar Siddhu

The method of loop invariants is used to prove ----- with respect to certain pre and post-conditions.

VuAnswers.com

Answer (Please select your correct option)

☐ correctness of a loop

correct

☐ correctness result

☐ correctness of variables

☐ correctness of algorithm

Made by: Waqar Siddhu

How many different ways can three of the letters of the word BYTES be chosen if the first letter must be B ?

VuAnswers.com

Answer (Please select your correct option)



$P(4,2)$



$P(2,4)$



$C(4,2)$

correct



None of these

Made by: Waqar Siddhu

The value of $0!$ is

VuAnswers.com

Answer (Please select your correct option)

correct

0

1

undefined

Made by: Waqar Siddhu

An arrangement of objects without the consideration of order is called

VuAnswers.com

Answer (Please select your correct option)



Permutation



Combination

correct



Selection



None of these

Made by: Waqar Siddhu

The value of ${}^nC_n =$

VuAnswers.com

Answer (Please select your correct option)

☐ 1

☐ 0

☐ n

☐ None of these

correct

Made by: Waqar Siddhu

If the order does not matter and repetition is allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)



$$n^k$$



$$C(n+k-1, k)$$

correct



$$P(n, k)$$



$$C(n, k)$$

Made by: Waqar Siddhu

If $(A \cup B) = A$ then

VuAnswers.com

Answer (Please select your correct option)

☐

$$(A \cap B) = B^c$$

☐

$$(A \cap B) = A$$

☐

$$(A \cap B) = B$$

correct

Made by: Waqar Siddhu

Among 200 people, 150 either play tennis or snooker or both. If 85 play tennis and 60 play tennis and snooker, how many play snooker?

Answer (Please select your correct option)

VuAnswers.com



25

correct



125



225



85

Made by: Waqar Siddhu

If A is a finite set then $n(A^c) =$

VuAnswers.com

Answer (Please select your correct option)



$$n(U) - n(A)$$

correct



$$n(U) + n(A)$$



$$n(A) - n(U)$$



$$0$$

Made by: Waqar Siddhu

What is the smallest integer N such that $\lceil N/5 \rceil = 7$?

VuAnswers.com

Answer (Please select your correct option)

☐ 35

correct

☐ 31

☐ 30

☐ 0

Made by: Waqar Siddhu

Compute $\lfloor -1.01 \rfloor$

Answer (Please select your correct option)

VuAnswers.com

☐

-2

☐

-1

correct

☐

2

☐

1

Made by: Waqar Siddhu

A procedure that yields a given set of possible outcomes is called

VuAnswers.com

Answer (Please select your correct option)

Event



Outcome



Experiment



correct

Made by: Waqar Siddhu

If a pair of dice is thrown then the probability of getting a total of 5 or 11 is

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{18}$

☐ $\frac{1}{9}$

☐ $\frac{1}{6}$

correct

Made by: Waqar Siddhu

The probability of getting a 5 when a die is thrown will be

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{6}$

correct

☐ $\frac{5}{6}$

☐ $\frac{1}{3}$

☐ $\frac{1}{36}$

Made by: Waqar Siddhu

If a die is rolled then what is the probability that the number is greater than 4?

VuAnswers.com

Answer (Please select your correct option)



$$\frac{1}{3}$$

correct



$$\frac{3}{4}$$



$$\frac{1}{2}$$



$$\frac{2}{3}$$

Made by: Waqar Siddhu

A rule that assigns a numerical value to each outcome in a sample space is called

VuAnswers.com

Answer (Please select your correct option)

☐ One to one function

☐ Conditional probability

☐ Random variable

correct

Made by: Waqar Siddhu

How many vertices will the graph have if it contain 16 edges and all vertices of degree 2?

VuAnswers.com

Answer (Please select your correct option)

☐ 14

☒ 16

☐ 18

☐ 32

correct

Made by: Waqar Siddhu

If a graph has any vertex of degree 3 then

VuAnswers.com

Answer (Please select your correct option)

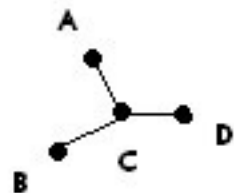
☐ It must have Euler circuit

☐ It must have Hamiltonian circuit

☐ It does not have Euler circuit

Made by: Waqar Siddhu

In the given graph, vertex "C" has degree



Answer (Please select your correct option)

VuAnswers.com

☐ 2

☒ 3

correct

☐ 4

☐ 6

Made by: Waqar Siddhu

The Common fraction for the recurring decimal $0.\overline{81}$ is

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{81}{100}$

☐ $\frac{81}{98}$

☐ $\frac{9}{11}$

☐ $\frac{81}{1000}$

correct

Made by: Waqar Siddhu

A sub graph of a graph G that contains every vertex of G and is a tree is called

VuAnswers.com

Answer (Please select your correct option)



Trivial tree



empty tree



Spanning tree

correct

Made by: Waqar Siddhu

If a tree has 8 vertices then it has

VuAnswers.com

Answer (Please select your correct option)

6 edges



7 edges



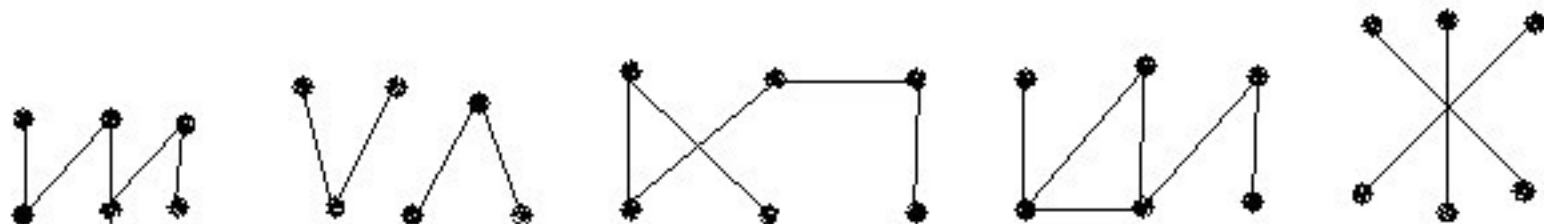
correct

9 edges



Made by: Waqar Siddhu

Which of the following graphs are tree?



Answer (Please select your correct option)

VuAnswers.com

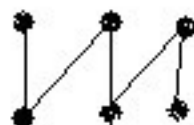
a, b, c

b, c, d

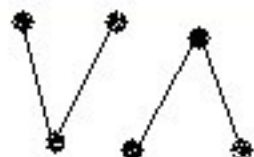
c, d, e

a, c, e

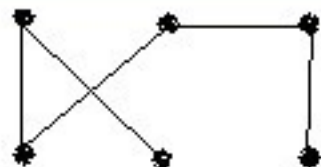
Made by: Waqar Siddhu



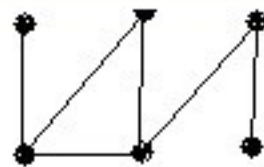
(a)



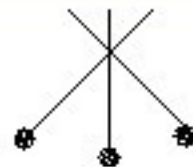
(b)



(c)



(d)



(e)

Answer (Please select your correct option)

VuAnswers.com



a, b, c



b, c, d



c, d, e



a, c, e

Made by: Waqar Siddhu

Every connected tree

Answer (Please select your correct option)

VuAnswers.com

☐

does not have spanning tree

☐

may or may not have spanning tree

☐

has a spanning tree

© Made by: Waqar Siddhu

Which of the following law is used to show?

$$p \leftrightarrow q \equiv q \leftrightarrow p$$

Answer (Please select your correct option)

VuAnswers.com

☐

Implication Law

☐

Commutative law

correct

☐

Exportation Law

☐

None of these

Made by: Waqar Siddhu

If p = It is red,

q = It is hot

Then "It is not red but hot" is denoted by : $p \wedge : q$.

Answer (Please select your correct option)

VuAnswers.com

True



correct

False



Made by: Waqar Siddhu

If p = It is red,

q = It is hot

Then "It is not red but hot" is denoted by : $p \wedge \neg q$.

Answer (Please select your correct option)

VuAnswers.com

True



False



Made by: Waqar Siddhu

A circuit with two input signals and one output signal is called

VuAnswers.com

Answer (Please select your correct option)



NOT-gate (or inverter)



AND- gate

correct



None of these

Made by: Waqar Siddhu

If A and B are two sets then the set of all elements that belong to A but not B is

VuAnswers.com

Answer (Please select your correct option)

☐

$$A \cup B$$

☐

$$A \cap B$$

☐

$$A - B$$

correct

☐

None of these

Made by: Waqar Siddhu

If $A = \{ \{5\} \}$, then power set of A is equal to

VuAnswers.com

Answer (Please select your correct option)

☐

$\{ \{ \emptyset, \{5\} \} \}$

☐

$\{ \emptyset, \{5\} \}$

correct

☐

$\{ \emptyset, \{ \{5\} \} \}$

☐

$\{ \{ \emptyset \}, \{5\} \}$

Made by: Waqar Siddhu

Let A be a set and R be a binary relation defined on it, if for all $t \in A, (t, t) \notin R$ then R is

VuAnswers.com

Answer (Please select your correct option)



Anti-symmetric



Symmetric



Irreflexive



Reflexive

Made by: Waqar Siddhu

Every relation _____

VuAnswers.com

Answer (Please select your correct option)

☐

may or may not be a function.

☐

is a bijective mapping.

☐

Cartesian product set

☐

is a function.

Made by: Waqar Siddhu

Let g be the function defined by $g(x) = 3x + 2$ then $g \circ g(x) =$

VuAnswers.com

Answer (Please select your correct option)

☐ $9x^2 + 4$

☐ $6x + 4$

☐ $9x + 8$

Made by: Waqar Siddhu

If $1+2+3+\dots+n = \frac{n(n+1)}{2}$ for all integers $n \geq 1$ then $P(k)$ is

VuAnswers.com

Answer (Please select your correct option)

☐ $1+2+3+\dots+k = \frac{k(k+1)}{2}$

☐ $1+2+3+\dots+n = \frac{n(n+1)}{2}$

☐ $1+2+3+\dots+(k+1) = \frac{(k+1)(k+2)}{2}$

correct

☐ $1+2+3+\dots+(k-1) = \frac{k(k-1)}{2}$

Made by: Waqar Siddhu

$n^2 > n + 3$ is true for all integers

VuAnswers.com

Answer (Please select your correct option)



$$n \geq 3$$

correct



$$n \geq 2$$



$$n \leq 3$$



$$n \geq 1$$

Made by: Waqar Siddhu

The direct proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)



considering q and then try to reach p



considering p and then try to reach q



considering p and $\sim q$ and try to reach contradiction



None of these

Made by: Waqar Siddhu

$n! > 2^n$ is true for all integers

VuAnswers.com

Answer (Please select your correct option)



$$n \geq 4$$



$$n \leq 4$$



$$n \geq 3$$



$$n \geq 2$$

Made by: Waqar Siddhu

An integer n is prime if and only if $n > 1$ and for all positive integers r and s , if $n = r.s$ then

VuAnswers.com

Answer (Please select your correct option)

correct

☐ $r = 1$ and $s = 2$

☐ $r = 1$ and $s = 0$

☐ $r = 2$ and $s = 3$

☐ None of these

Made by: Waqar Siddhu

What is the contra positive of the given statement:

If square root of every prime number is irrational then square root of 2 is irrational.

Answer (Please select your correct option)

VuAnswers.com

☐

If square root of 2 is not irrational then square root of every prime number is not irrational.

correct

☐

If square root of 2 is irrational then square root of every prime number is not irrational.

☐

If square root of 2 is not irrational then square root of every prime number is irrational.

☐

If square root of every prime number is not irrational then square root of 2 is not irrational.

Made by: Waqar Siddhu

The word ----- refers to a step-by-step method for performing some action.

Answer (Please select your correct option)

VuAnswers.com

☐

Series

☐

Relation

☐

Algorithm

correct

☐

Function

Made by: Waqar Siddhu

If r is a positive integer then $\gcd(r, 0) =$

VuAnswers.com

Answer (Please select your correct option)

☐

None of these

☐

r

☐

5

☐

0

Made by: Waqar Siddhu

Suppose that there are eight runners in a race first will get gold medal, the second will get silver and third will get bronze. How many different ways are there to award these medals if all possible outcomes of race can occur and there is no tie.

Answer (Please select your correct option)

VuAnswers.com

☐

$P(8,3)$

☐

$P(100,97)$

☐

$P(97,3)$

☐

None of these

Made by: Waqar Siddhu

The value of $\frac{(n-1)!}{(n+1)!}$ is

VuAnswers.com

Answer (Please select your correct option)

☐

0

☐

$n(n-1)$

☐

$\frac{1}{(n^2+n)}$

☐

can not be determined

Made by: Waqar Siddhu

An arrangement of objects without the consideration of order is called

VuAnswers.com

Answer (Please select your correct option)

☐

Permutation

☐

Combination

☐

Selection

☐

None of these

Made by: Waqar Siddhu

The same element can never appear ----- in a set.

Answer (Please select your correct option)

VuAnswers.com

☐ twice

☐ once

☐ thrice

Made by: Waqar Siddhu

To find the number of unordered partitions, we have to count the ----- partitions and then divide it by suitable number to erase the order in partitions.

Answer (Please select your correct option)

VuAnswers.com



unordered



ordered

correct



random



None of these

Made by: Waqar Siddhu

If A and B are two disjoint sets then which of the following must be true

VuAnswers.com

Answer (Please select your correct option)



$$n(A \cup B) = n(A) + n(B)$$



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$



$$n(A \cup B) = \phi$$



None of these

Made by: Waqar Siddhu

If A is a finite set then $n(A^c) =$

VuAnswers.com

Answer (Please select your correct option)

☐

$$n(U) - n(A)$$

correct

☐

$$n(U) + n(A)$$

☐

$$n(A) - n(U)$$

☐

$$0$$

Made by: Waqar Siddhu

The value of $\lceil x \rceil$ for $x = -2.01$ is

VuAnswers.com

Answer (Please select your correct option)

☐

-3

☐

1

☐

-2

☐

0

Made by: Waqar Siddhu

If a pair of dice is thrown then the probability of getting a total of 5 or 11 is

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{18}$

☐ $\frac{1}{9}$

☐ $\frac{1}{6}$

Made by: Waqar Siddhu

A die is thrown twice. What is the probability that the sum of the number of dots shown is 3 or 11?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{2}{3}$

☐ $\frac{1}{9}$

☐ $\frac{1}{2}$

Made by: Waqar Siddhu

If A and B are independent events then $P(A/B) =$

VuAnswers.com

Answer (Please select your correct option)

☐

$P(B)$

☐

$P(A)$

☐

$P(A \cap B)$

☐

$P(A \cup B)$

Made by: Waqar Siddhu

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

VuAnswers.com

Answer (Please select your correct option)

☐ Dependent

☐ Independent

☐ Exclusive

Made by: Waqar Siddhu

If A, B and C are any three events, then $P(A \cup B \cup C)$ is equal to

VuAnswers.com

Answer (Please select your correct option)

☐

$$P(A) + P(B) + P(C)$$

☐

$$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

☐

$$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$$

☐

$$P(A) + P(B) + P(C) + P(A \cap B \cap C)$$

Made by: Waqar Siddhu

The expectation of x is equal to

VuAnswers.com

Answer (Please select your correct option)

☐

Sum of all terms

☐

Sum of all terms divided by number of terms

☐

$\sum xf(x)$

Made by: Waqar Siddhu

The degrees of the vertices a, b, c, d, e respectively, in the given graph are



Answer (Please select your correct option)

VuAnswers.com

☐

2, 2, 3, 1, 1

☐

2, 3, 1, 0, 1

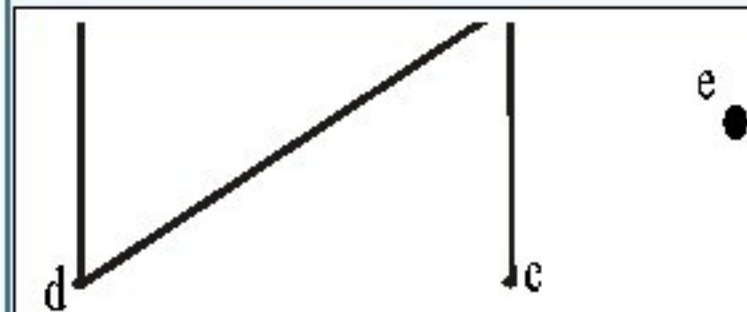
☐

0, 1, 2, 2, 0

☐

2, 3, 1, 2, 0

Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

☐

2, 2, 3, 1, 1

☐

2, 3, 1, 0, 1

☐

0, 1, 2, 2, 0

☐

2, 3, 1, 2, 0

Made by: Waqar Siddhu

The Hamiltonian circuit for the following graph is



Answer (Please select your correct option)

VuAnswers.com

☐ abcdefgh

☐ abefgha

☐ abcdefgha

Made by: Waqar Siddhu

Two matrices are said to be conformable for multiplication if number of(a)..... of 1st matrix is equal to number of(b)..... in 2nd matrix

Answer (Please select your correct option)

VuAnswers.com



(a) rows, (b) columns



(a) columns, (b) rows



(a) columns, (b) columns



(a) rows, (b) rows

Made by: Waqar Siddhu

The list of the degrees of the vertices of a graph in non-increasing order is called

VuAnswers.com

Answer (Please select your correct option)



Isomorphic Invariant



Degree Sequence



Order of Graph



Lenght of Circuit

Made by: Waqar Siddhu

Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does the graph have?

Answer (Please select your correct option)

VuAnswers.com

☐

12

☐

13

☐

14

☐

8

Made by: Waqar Siddhu

How many non-isomorphic spanning trees does the following simple graph has?



VuAnswers.com

Answer (Please select your correct option)

6

☐

7

☐

8

☒

correct

Made by: Waqar Siddhu

What is the output state of an AND gate if the inputs are 0 and 1?

VuAnswers.com

Answer (Please select your correct option)

☐ 0

correct

☐ 1

☐ 2

☐ 3

Made by: Waqar Siddhu

Every connected tree

VuAnswers.com

Answer (Please select your correct option)

does not have spanning tree

☐

may or may not have spanning tree

☐

has a spanning tree

☐

correct

Made by: Waqar Siddhu

The following directed graph can be represented in form of matrix as



VuAnswers.com

Answer (Please select your correct option)

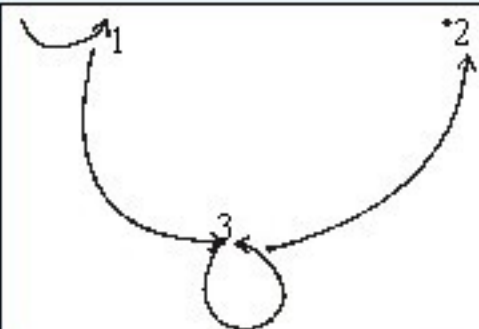
☐
$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \cdot \end{pmatrix}$$

☐
$$\begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \wedge \end{pmatrix}$$

☐
$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \wedge \end{pmatrix}$$

☐
$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \wedge & \cdot \end{pmatrix}$$

Made by: Waqar Siddhu



VuAnswers.com

Answer (Please select your correct option)

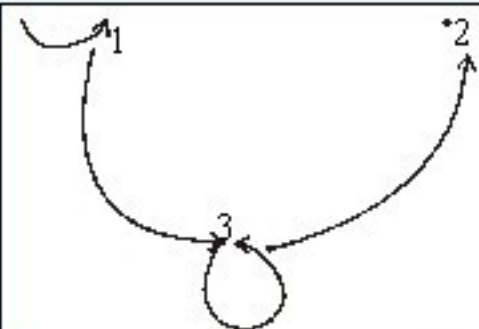
☐
$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \cdot \end{pmatrix}$$

☐
$$\begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \wedge \end{pmatrix}$$

☐
$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ \wedge & \cdot & \wedge \end{pmatrix}$$

☐
$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ \wedge & \wedge & \cdot \end{pmatrix}$$

Made by: Waqar Siddhu



VuAnswers.com

Answer (Please select your correct option)

☐
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

☐
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

☐
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

☐
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Made by: Waqar Siddhu

Rephrase the following statement in bi-conditional form
"If you get up early in the morning, you will be healthy"

VuAnswers.com

Answer (Please select your correct option)

☐ You will be healthy if and only if you get up early in the morning

correct

☐ If you will be healthy then you will get up early in the morning

☐ If you will get up early in the morning then you will be healthy

☐ None of these

Made by: Waqar Siddhu

An argument is _____ if the conclusion is true when all the premises are true.

VuAnswers.com

Answer (Please select your correct option)

☐ Invalid

☐ False

☐ Valid

☐ None of these

correct

Made by: Waqar Siddhu

How many input signals are required for an OR-gate?

VuAnswers.com

Answer (Please select your correct option)

- ☐ 2 correct
- ☐ 8
- ☐ All multiples of two
- ☐ 1

Made by: Waqar Siddhu

What will be the output of an OR-gate if it has inputs 0 and 1?

VuAnswers.com

Answer (Please select your correct option)

- ☐ 0
- ☒ 1
- ☐ 2
- ☐ 3

correct

Made by: Waqar Siddhu

Let f is defined recursively by $f(0) = 3, f(n+1) = 2f(n) + 3$ then $f(1) =$

VuAnswers.com

Answer (Please select your correct option)

☒ 9 correct

☐ 10

☐ 18

☐ 21

Made by: Waqar Siddhu

Let a and b be integers. Suppose a function Q is defined recursively as follows:

$$Q(a,b) = \begin{cases} 5 & \text{if } a < b \\ Q(a-b, b+2) + a & \text{if } b \leq a \end{cases}$$

Then $Q(2,7) =$

VuAnswers.com

Answer (Please select your correct option)

☐

7

☐

2

☐

5

☐

0

Made by: Waqar Siddhu

$P(n)$ is called statement or

VuAnswers.com

Answer (Please select your correct option)



sentence



proposition



inequality



none of these

correct

Made by: Waqar Siddhu

The direct proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)

- ☐ considering q and then try to reach p
- ☐ considering p and then try to reach q
- ☐ considering p and $\sim q$ and try to reach contradiction
- ☐ None of these

correct

Made by: Waqar Siddhu

The indirect proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)

- ☐ Considering : q and then try to reach p
- ☐ Considering p and : q are true and try to reach contradiction
- ☐ Considering p and then try to reach q
- ☐ Considering : p and then try to reach q

correct

Made by: Waqar Siddhu

The contradiction proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)

- ☐ Considering p and then try to reach q
- ☒ Considering $\neg q$ and then try to reach $\neg p$
- ☐ Considering p and $\neg q$ are true and try to reach contradiction
- ☐ None of these

correct

Made by: Waqar Siddhu

$n! > 2^n$ is true for all integers

VuAnswers.com

Answer (Please select your correct option)

☐ $n \geq 4$

☐ $n \leq 4$

☐ $n \geq 3$

☐ $n \geq 2$

Made by: Waqar Siddhu

An integer n is prime if and only if $n > 1$ and for all positive integers r and s , if $n = r.s$ then

VuAnswers.com

Answer (Please select your correct option)

☐ $r = 1$ and $s = 2$

correct

☐ $r = 1$ and $s = 0$

☐ $r = 2$ and $s = 3$

☐ None of these

Made by: Waqar Siddhu

The word ----- refers to a step-by-step method for performing some action.

VuAnswers.com

Answer (Please select your correct option)

☐ Series

☐ Relation

☐ Algorithm

☐ Function

correct

Made by: Waqar Siddhu

$\gcd(4,12) =$

VuAnswers.com

Answer (Please select your correct option)

- ☐ 2
- ☐ 3
- ☒ 4
- ☐ 12
- correct

Made by: Waqar Siddhu

How many ways are there to select five players from a 10 members tennis team to arrange a match with another school?

VuAnswers.com

Answer (Please select your correct option)

☐ $C(10,5)$

correct

☐ $C(5,10)$

☐ $P(10,5)$

☐ None of these

Made by: Waqar Siddhu

The value of $\frac{n!}{(n-1)!}$ is

VuAnswers.com

Answer (Please select your correct option)

- ☐ n correct
- ☐ $n-1$
- ☐ 0
- ☐ can not be determined

Made by: Waqar Siddhu

The number of k -combinations that can be chosen from a set of n elements can be written as

VuAnswers.com

Answer (Please select your correct option)

☐

nC_k

☐

kC_n

☐

nP_k

☐

kP_n

Made by: Waqar Siddhu

If the order matters and repetition is not allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)

☐

$$n^k$$

☐

$$C(n+k-1, k)$$

☐

$$P(n, k)$$

☐

$$C(n, k)$$

Made by: Waqar Siddhu

When $A \subseteq B$, then B is called a ----- of A.

VuAnswers.com

Answer (Please select your correct option)

☐

superset

correct

☐

subset

☐

power set

☐

none of these

Made by: Waqar Siddhu

Null set is denoted by

VuAnswers.com

Answer (Please select your correct option)

☐

ϕ or { }

correct

☐

A

☐

N

Made by: Waqar Siddhu

Among 200 people, 150 either play tennis or snooker or both. If 85 play tennis and 60 play tennis and snooker, how many play snooker?

VuAnswers.com

Answer (Please select your correct option)

☐ 125

☐ 225

☐ 85

☐ 25

correct

Made by: Waqar Siddhu

If A is a finite set then $n(A^c) =$

VuAnswers.com

Answer (Please select your correct option)

☐ $n(U) - n(A)$

correct

☐ $n(U) + n(A)$

☐ $n(A) - n(U)$

☐ 0

Made by: Waqar Siddhu

The value of $\lceil x \rceil$ for $x = -3.01$ is

VuAnswers.com

Answer (Please select your correct option)

☐ -3.01

☒ -3

correct

☐ -2

☐ -1.99

Made by: Waqar Siddhu

What is the probability of getting a number greater than 4 when a die is thrown?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{2}$

☐ $\frac{3}{2}$

☐ $\frac{1}{3}$

correct

☐ 1

Made by: Waqar Siddhu

If two fair dice are thrown, what is the probability of getting a double six?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{6}$

☐ $\frac{1}{36}$

correct

☐ 1

☐ $\frac{1}{2}$

Made by: Waqar Siddhu

A die is thrown twice. What is the probability that the sum of the number of dots shown is 3 or 11?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{2}{3}$

☐ $\frac{1}{9}$

correct

☐ $\frac{1}{2}$

Made by: Waqar Siddhu

A rule that assigns a numerical value to each outcome in a sample space is called

VuAnswers.com

Answer (Please select your correct option)

☐ One to one function

☐ Conditional probability

☐ Random variable

correct

Made by: Waqar Siddhu

The expectation μ for the following table is

x_i	1	3
$f(x_i)$	0.4	0.1

VuAnswers.com

Answer (Please select your correct option)

☐ 0.5

☐ 3.4

☐ 0.3

☐ 0.7

Made by: Waqar Siddhu

How many vertices will the graph have if it contain 16 edges and all vertices of degree 2?

VuAnswers.com

Answer (Please select your correct option)

☐ 14

☒ 16

☐ 18

☐ 32

correct

Made by: Waqar Siddhu

Two distinct edges with the same set of end points are called

VuAnswers.com

Answer (Please select your correct option)

Isolated

☐

Incident

☐

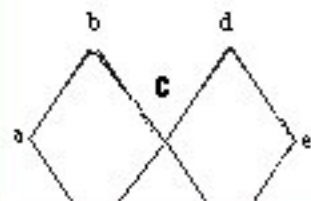
Parallel

☐

correct

Made by: Waqar Siddhu

The graph given below



Answer (Please select your correct option)

VuAnswers.com

has not Euler circuit

☐

has Hamiltonian circuit

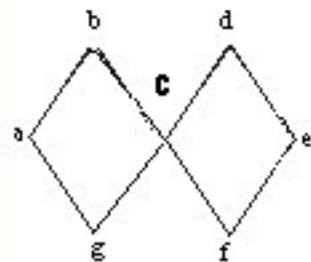
☐

does not have Hamiltonian circuit

☐

correct

Made by: Waqar Siddhu



VuAnswers.com

Answer (Please select your correct option)

☐ has not Euler circuit☐ has Hamiltonian circuit☐ does not have Hamiltonian circuit

Made by: Waqar Siddhu

Changing rows of a matrix into its columns is called

VuAnswers.com

Answer (Please select your correct option)

☐ symmetric matrix

☐ transpose of matrix

☐ adjoint of matrix

☐ Hermitian Matrix

correct

Made by: Waqar Siddhu

If the transpose of any square matrix is equal to the matrix itself then it is called

VuAnswers.com

Answer (Please select your correct option)

☐ Additive Inverse

☐ Hermitian Matrix

☐ Symmetric Matrix

☐ Singular Matrix

Made by: Waqar Siddhu

The given graph is called a ----- graph.



VuAnswers.com

Answer (Please select your correct option)

Simple



Complete



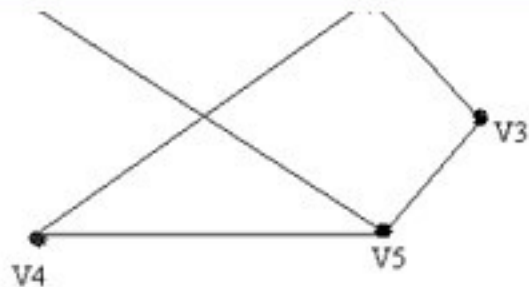
Complete Bipartite



Non-planar



Made by: Waqar Siddhu



VuAnswers.com

Answer (Please select your correct option)

- ☐ Simple
- ☐ Complete
- ☐ Complete Bipartite
- ☐ Non-planar

Made by: Waqar Siddhu

In a planar graph, the number of crossing edges are

VuAnswers.com

Answer (Please select your correct option)

☐ 0

☐ 1

☐ 2

☐ 3

Made by: Waqar Siddhu

If T is a full binary tree and has 5 internal vertices then the total vertices of T are

VuAnswers.com

Answer (Please select your correct option)

☐ 11

☐ 12

☐ 13

☐ 10

Made by: Waqar Siddhu

A sub graph of a graph G that contains every vertex of G and is a tree is called

Answer (Please select your correct option)

VuAnswers.com

Trivial tree



empty tree



Spanning tree



correct

Made by: Waqar Siddhu

A circuit that consist of a single vertex is called

Answer (Please select your correct option)

VuAnswers.com

☐ Tree

☐ Empty

☐ Trivial

correct

Made by: Waqar Siddhu

A vertex of degree 1 in a tree is called

Answer (Please select your correct option)

VuAnswers.com

☐ Internal vertex

☒ Terminal vertex

☐ Sibling vertex

correct

Made by: Waqar Siddhu

Complete graph is planar if

Answer (Please select your correct option)

VuAnswers.com

☐ $n = 4$

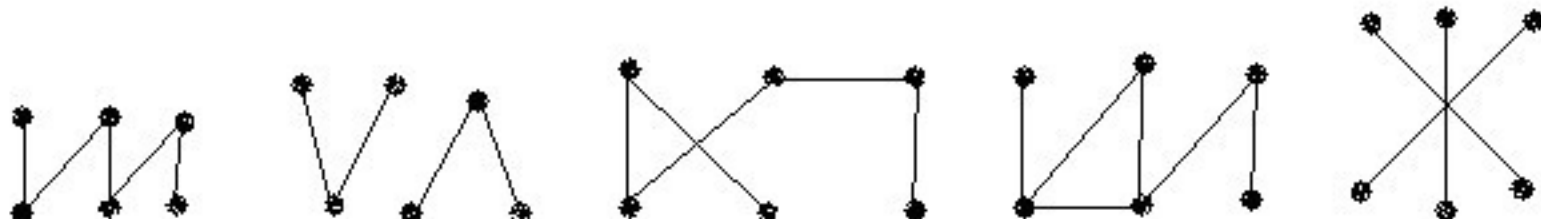
☐ $n > 4$

☐ $n \leq 4$

correct

Made by: Waqar Siddhu

Which of the following graphs are tree?



Answer (Please select your correct option)

VuAnswers.com

a, b, c

b, c, d

c, d, e

a, c, e

Made by: Waqar Siddhu

Euler formula for graphs is

Answer (Please select your correct option)

VuAnswers.com

☐ $f = e - v$

☐ $f = e + v + 2$

☐ $f = e - v - 2$

☐ $f = e - v + 2$

correct

Made by: Waqar Siddhu

The logical expression $p \vee q$ will be read as

Answer (Please select your correct option)

VuAnswers.com

☐ $p \text{ or } q$

correct

☐ $p \text{ and } q$

☐ $p \times q$

☐ $p - q$

Made by: Waqar Siddhu

In method of proof by contradiction, we suppose the statement to be proved is false.

Answer (Please select your correct option)

VuAnswers.com

☐ True

☐ False

Made by: Waqar Siddhu

Let U be the universal set and A is its subset then $A \cap A^c$ is equal to

Answer (Please select your correct option)

VuAnswers.com

☐

A

☐

A^c

☐

ϕ

correct

☐

U

Made by: Waqar Siddhu

If A and S are two reflexive relations then $A \cap S$ will be

Answer (Please select your correct option)

VuAnswers.com

☐ Symmetric

☐ Reflexive

correct

☐ Transitive

Made by: Waqar Siddhu

If two relations are reflexive then their composition is

Answer (Please select your correct option)

VuAnswers.com



Anti-symmetric



Reflexive

correct



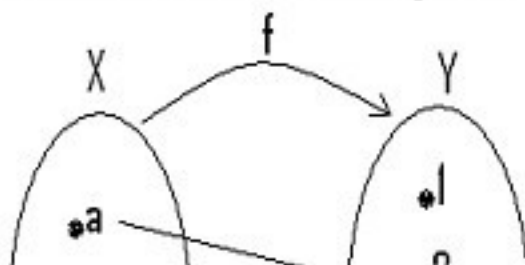
Irreflexive



Symmetric

Made by: Waqar Siddhu

If $X = \{a, b, c\}$ and $Y = \{1, 2, 3, 4\}$. Let us define a function $f : X \rightarrow Y$ as shown in following figure then the inverse image of 1 is



Answer (Please select your correct option)

VuAnswers.com

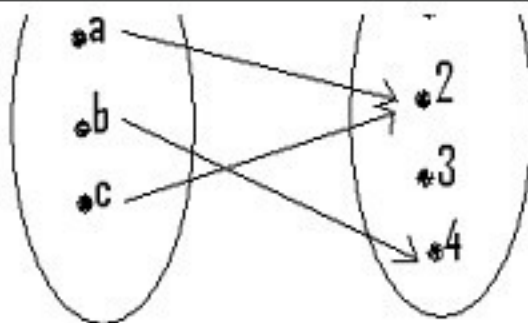
☐ 1

☐ 2

☐ 1,2

☐ None of these

Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

☐ 1☐ 2☐ 1,2☐ None of these

Made by: Waqar Siddhu

Let f and g be the functions defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$ then composition of f and g is

Answer (Please select your correct option)

VuAnswers.com

☐ $6x + 6$

☐ $5x + 5$

☐ $6x + 7$

correct

Made by: Waqar Siddhu

The composition of two functions is _____.

Answer (Please select your correct option)

VuAnswers.com

☐ One to one

☐ Bijective

☐ Not commutative

correct

Made by: Waqar Siddhu

Let f is defined recursively by $f(0) = 5, f(n+1) = 4f(n) + 2$ then $f(1) =$

Answer (Please select your correct option)

VuAnswers.com

8



10



21



22



correct

Made by: Waqar Siddhu

$P(n)$ is called statement or

Answer (Please select your correct option)

VuAnswers.com

☐ sentence

☐ proposition

☐ inequality

☐ none of these

correct

Made by: Waqar Siddhu

Let n and d be the integers and $d \neq 0$. Then n is divisible by d or d divides n iff

Answer (Please select your correct option)

VuAnswers.com

☐

$n = k.d$ for some integer k .

correct

☐

$n - k = d$

☐

$n.d = 1$

☐

$n.k = 0$

Made by: Waqar Siddhu

The indirect proof of a statement $p \rightarrow q$ involves

Answer (Please select your correct option)

VuAnswers.com



Considering : q and then try to reach p



Considering p and : q are true and try to reach contradiction

correct



Considering p and then try to reach q



Considering : p and then try to reach q

Made by: Waqar Siddhu

The contra positive proof of a statement $p \rightarrow q$ involves

Answer (Please select your correct option)

VuAnswers.com



Considering p and then try to reach q



Considering : q and then try to reach : p

correct



Considering p and : q and try to reach contradiction



None of these

Made by: Waqar Siddhu

For all positive integer values of n , $5^n - 1$ is divisible by

Answer (Please select your correct option)

VuAnswers.com

3



4



correct

6



0



Made by: Waqar Siddhu

An integer n is prime if and only if $n > 1$ and for all positive integers r and s , if $n = r.s$ then

Answer (Please select your correct option)

VuAnswers.com



$r = 1$ and $s = 2$



$r = 1$ and $s = 0$



$r = 2$ and $s = 3$



None of these

Made by: Waqar Siddhu

An integer n is odd for some integer k iff

Answer (Please select your correct option)

VuAnswers.com



$$n = 2k$$



$$n = 2(k+1)$$



$$n = 2(k-1)$$



$$n = 2k+1$$

correct

Made by: Waqar Siddhu

If a and b are any positive integers with $b \neq 0$ and q and r are non negative integers such that $a = b \cdot q + r$ then

Answer (Please select your correct option)

VuAnswers.com

☐ $\gcd(a, b) = \gcd(b, r)$

☐ $\gcd(a, r) = \gcd(b, r)$

☐ $\gcd(a, q) = \gcd(q, r)$

Made by: Waqar Siddhu

The greatest common divisor of 5 and 10 is

Answer (Please select your correct option)

VuAnswers.com

☐ 5

correct

☐ 0

☐ 1

☐ None of these

Made by: Waqar Siddhu

How many ways are there to select a first prize winner, a second prize winner and a third prize winner from 100 different people who have entered in a contest.

Answer (Please select your correct option)

VuAnswers.com



$P(97, 3)$



$P(100, 3)$



$P(100, 97)$



None of these

Made by: Waqar Siddhu

If one event can occur in n_1 ways, a second event can occur in n_2 (different) ways, then the total number of ways in which exactly one of the events (i.e., first or second) can occur is

Answer (Please select your correct option)

VuAnswers.com

☐ $n_1 + n_2$

☐ $n_1 n_2$

☐ $2n_1 n_2$

☐ $2^{n_1 n_2}$

Made by: Waqar Siddhu

The value of $\frac{(n+1)!}{(n-1)!}$ is

Answer (Please select your correct option)

VuAnswers.com

☐

0

☐

$n(n-1)$

☐

$n^2 + n$

☐

can not be determined

Made by: Waqar Siddhu

If $(A \cup B) = A$ then

Answer (Please select your correct option)

VuAnswers.com

☐

$$(A \cap B) = B^c$$

☐

$$(A \cap B) = A$$

☐

$$(A \cap B) = B$$

correct

Made by: Waqar Siddhu

To find the number of unordered partitions, we have to count the ----- partitions and then divide it by suitable number to erase the order in partitions.

Answer (Please select your correct option)

VuAnswers.com

☐ unordered

☐ ordered

☐ random

☐ None of these

correct

Made by: Waqar Siddhu

If A and B are finite (overlapping) sets, then which of the following must be true

Answer (Please select your correct option)

VuAnswers.com

☐

$$n(A \cup B) = n(A) + n(B)$$

☐

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

correct

☐

$$n(A \cup B) = \phi$$

☐

None of these

Made by: Waqar Siddhu

What is the smallest integer N such that $\left\lceil \frac{N}{6} \right\rceil = 9$

Answer (Please select your correct option)

VuAnswers.com

☐ 46

☐ 29

☐ 49

☐ 64

correct

Made by: Waqar Siddhu

A procedure that yields a given set of possible outcomes is called

Answer (Please select your correct option)

VuAnswers.com

☐ Event

☐ Outcome

☐ Experiment

correct

Made by: Waqar Siddhu

What is the probability of getting a number greater than 4 when a die is thrown?

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{1}{2}$

☐ $\frac{3}{2}$

☐ $\frac{1}{3}$

☐ 1

correct

Made by: Waqar Siddhu

The ----- of the experiment is the set of possible outcomes.

Answer (Please select your correct option)

VuAnswers.com

event

☐

sample space

☐

correct

subset

☐

Made by: Waqar Siddhu

If two fair dice are thrown, what is the probability of getting a double six?

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{1}{6}$ correct

☐ $\frac{1}{36}$

☐ 1

☐ $\frac{1}{2}$

Made by: Waqar Siddhu

If a die is thrown then the probability that the dots on the top are prime numbers or odd numbers is

Answer (Please select your correct option)

VuAnswers.com

1

☐

$\frac{1}{3}$

☐

$\frac{2}{3}$

☒

correct

Made by: Waqar Siddhu

The expectation μ for the following table is

x_i	1	3
$f(x_i)$	0.4	0.1

Answer (Please select your correct option)

VuAnswers.com

☐

0.5

☐

3.4

☐

0.3

☐

0.7

Made by: Waqar Siddhu

A line segment joining pair of vertices is called

Answer (Please select your correct option)

VuAnswers.com

☐ Loop

☐ Edge

☐ Node

correct

Made by: Waqar Siddhu

Changing rows of a matrix into its columns is called

Answer (Please select your correct option)

VuAnswers.com

☐ symmetric matrix

☐ transpose of matrix

☐ adjoint of matrix

☐ Hermitian Matrix

correct

Made by: Waqar Siddhu

The given graph is called a ----- graph.



Answer (Please select your correct option)

VuAnswers.com

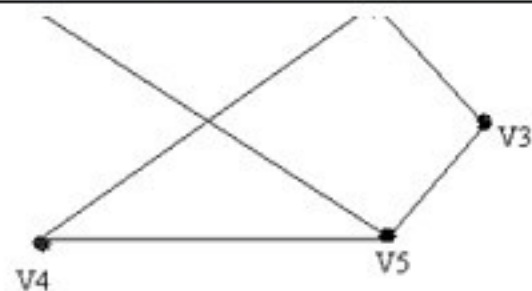
☐ Simple

☐ Complete

☐ Complete Bipartite

☐ Non-planar

Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

☐ Simple☐ Complete☐ Complete Bipartite☐ Non-planar

Made by: Waqar Siddhu

A sub graph of a graph G that contains every vertex of G and is a tree is called

VuAnswers.com

Answer (Please select your correct option)

Trivial tree



empty tree



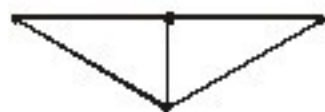
Spanning tree



correct

Made by: Waqar Siddhu

How many non-isomorphic spanning trees does the following simple graph has?



Answer (Please select your correct option)

VuAnswers.com

6

☐

7

☐

8

☐

correct

Made by: Waqar Siddhu

A vertex of degree greater than 1 in a tree is called a

VuAnswers.com

Answer (Please select your correct option)

Branch vertex



correct

Terminal vertex

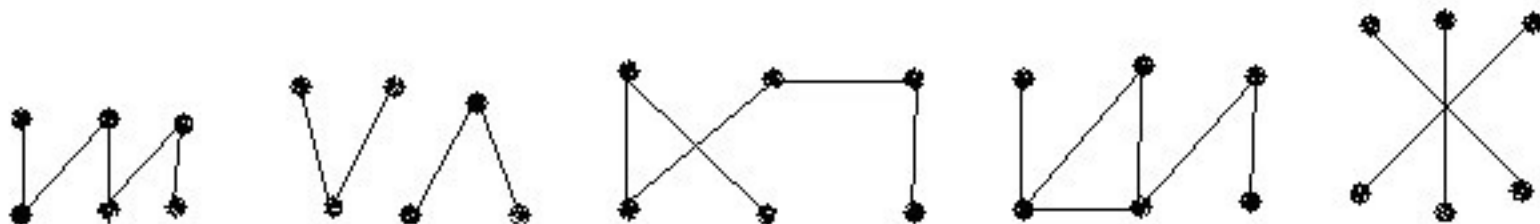


Ancestor



Made by: Waqar Siddhu

Which of the following graphs are tree?



Answer (Please select your correct option)

VuAnswers.com

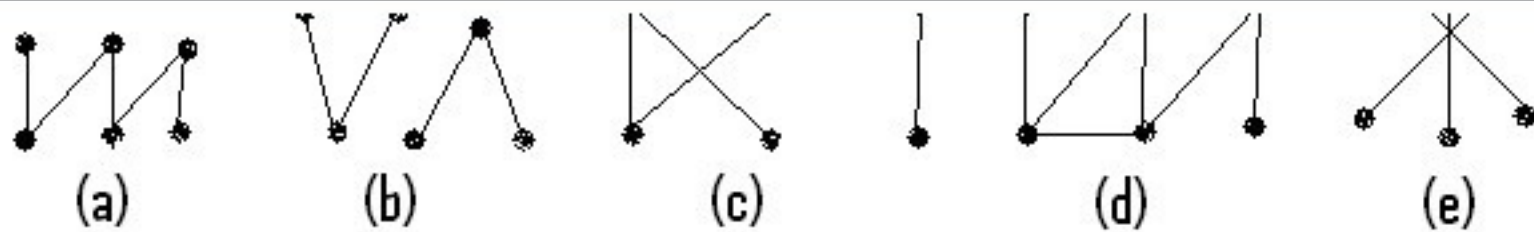
a, b, c

b, c, d

c, d, e

a, c, e

Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

☐ a, b, c☐ b, c, d☐ c, d, e☐ a, c, e

Made by: Waqar Siddhu

Any two spanning trees for a graph

VuAnswers.com

Answer (Please select your correct option)

- ☐ Does not contain same number of edges
- ☐ Have the same degree of corresponding edges
- ☒ contain same number of edges
- ☐ May or may not contain same number of edges

correct

Made by: Waqar Siddhu

If p & q are statements, then their disjunction is

VuAnswers.com

Answer (Please select your correct option)

☐ p or q

correct

☐ p and q

☐ p and q and p

☐ None of these

Made by: Waqar Siddhu

If p and q are statement variables then bi-conditional of p and q is denoted by

VuAnswers.com

Answer (Please select your correct option)

☐ $p \leftrightarrow q$

correct

☐ $p \rightarrow q$

☐ $q \rightarrow p$

☐ None of these

Made by: Waqar Siddhu

How many input signals are required for an OR-gate?

VuAnswers.com

Answer (Please select your correct option)

☐ 2

correct

☐ 8

☐ All multiples of two

☐ 1

Made by: Waqar Siddhu

A set which contains no element is called a

VuAnswers.com

Answer (Please select your correct option)

Null set



correct

Universal set



None of these



Made by: Waqar Siddhu

If A and B are two sets then the set of all elements that belong to A but not B is

VuAnswers.com

Answer (Please select your correct option)

☐ $A \cup B$

☐ $A \cap B$

☐ $A - B$

☐ None of these

correct

Made by: Waqar Siddhu

Inverse of relation can be obtained by

VuAnswers.com

Answer (Please select your correct option)

☐ changing signs of elements in order pairs.

☐ changing position of elements in order pairs.

☐ taking multiplicative inverse of elements in order pairs.

correct

not sure

Made by: Waqar Siddhu

If $f(x) = 2x + 1$ and $g(x) = x^2 - 1$ then $g \circ g(x) =$

VuAnswers.com

Answer (Please select your correct option)

☐ $x^3 - 2x^2$

☐ $x^4 - 2x^2$

correct

☐ $x^4 + 1$

☐ $x^4 + 2x^2$

Made by: Waqar Siddhu

If f and g are two one-to-one functions then their composition that is $f \circ g$ is

VuAnswers.com

Answer (Please select your correct option)

☐ Not One-to-One

☐ On to

☐ One-to-One

☐ One-to-One and Onto

Made by: Waqar Siddhu

If the n th term of a sequence is $a_n = 2(-3)^n + 5^n$ then the term a_1 is

VuAnswers.com

Answer (Please select your correct option)

-1



correct

0



1



2



Made by: Waqar Siddhu

When 5^k is even, then $5^k + 5^k + 5^k$ must be

VuAnswers.com

Answer (Please select your correct option)

correct

even

odd

negative

none of these

Made by: Waqar Siddhu

Proof of a statement by induction comprises of two basic steps:

VuAnswers.com

Answer (Please select your correct option)

☐ Inductive and Deductive

☐ Basis and Inductive

☐ Arranging and Sorting

☐ None of these

correct

Made by: Waqar Siddhu

In Mathematical Induction, inductive step is

VuAnswers.com

Answer (Please select your correct option)



$$\forall k, P(k) \rightarrow P(n)$$



$$\forall k, P(k) \rightarrow P(k+1)$$

correct



$$\forall k, P(k) \rightarrow P(n+1)$$



$$\forall k, P(k) \rightarrow P(k-1)$$

Made by: Waqar Siddhu

$n^2 > n + 3$ is true for all integers

VuAnswers.com

Answer (Please select your correct option)



$$n \geq 3$$

correct



$$n \geq 2$$



$$n \leq 3$$



$$n \geq 1$$

Made by: Waqar Siddhu

For all positive integers $n^2 + n$ is divisible by

VuAnswers.com

Answer (Please select your correct option)

3



2



correct

7



5



Made by: Waqar Siddhu

The contradiction proof of a statement $p \rightarrow q$ involves

VuAnswers.com

Answer (Please select your correct option)

- ☐ Considering p and then try to reach q
- ☐ Considering : q and then try to reach : p
- ☐ Considering p and : q are true and try to reach contradiction
- ☐ None of these

correct

not totaly sure

Made by: Waqar Siddhu

The sum of two irrational numbers must be

VuAnswers.com

Answer (Please select your correct option)

- ☐ irrational number
- ☐ rational number
- ☐ integer
- ☐ depends on numbers which are added

correct

Made by: Waqar Siddhu

The greatest common divisor of 5 and 10 is

VuAnswers.com

Answer (Please select your correct option)

☐ 5

correct

☐ 0

☐ 1

☐ None of these

Made by: Waqar Siddhu

If $n(A_i)$ denotes the number of elements in set A_i then $n(A_1 \cup A_2 \cup \dots \cup A_m) = n(A_1) + n(A_2) + \dots + n(A_m)$ where

VuAnswers.com

Answer (Please select your correct option)

☐

$$A_i \cup A_j = \phi \text{ if } i = j$$

☐

$$A_i \cap A_j = \phi \text{ if } i = j$$

☐

$$A_i \cup A_j = \phi \text{ if } i \neq j$$

☐

$$A_i \cap A_j = \phi \text{ if } i \neq j$$

correct

Made by: Waqar Siddhu

An arrangement of objects with the consideration of order is called

VuAnswers.com

Answer (Please select your correct option)

☐ Permutation

☐ Combination

correct

80% sure

☐ Selection

☐ None of these

Made by: Waqar Siddhu

If the order matters and repetition is allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)

☐ $C(n, k)$

☐ n^k

correct

☐ $C(n + k - 1, k)$

☐ $P(n, k)$

Made by: Waqar Siddhu

If the order matters and repetition is not allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)

☐ n^k

☐ $C(n+k-1, k)$

☐ $P(n, k)$

correct

☐ $C(n, k)$

Made by: Waqar Siddhu

If the order does not matter and repetition is not allowed then total number of ways for selecting k sample from n number of elements is

VuAnswers.com

Answer (Please select your correct option)

☐ $P(n, k)$

☐ $C(n, k)$

correct

☐ n^k

☐ $C(n + k - 1, k)$

Made by: Waqar Siddhu

The same element can never appear ----- in a set.

VuAnswers.com

Answer (Please select your correct option)

twice

☐

correct

once

☐

thrice

☐

Made by: Waqar Siddhu

If A is a finite set then $n(A^c) =$

VuAnswers.com

Answer (Please select your correct option)

☐ $n(U) - n(A)$

correct

☐ $n(U) + n(A)$

☐ $n(A) - n(U)$

☐ 0

Made by: Waqar Siddhu

Compute $\lfloor -1.01 \rfloor$

Answer (Please select your correct option)

VuAnswers.com

☐ -2

☒ -1

correct

☐ 2

☐ 1

Made by: Waqar Siddhu

What is the probability of getting a number greater than 4 when a die is thrown?

VuAnswers.com

Answer (Please select your correct option)

☐ $\frac{1}{2}$

☐ $\frac{3}{2}$

☐ $\frac{1}{3}$

☐ 1

correct

Made by: Waqar Siddhu

The ----- of the experiment is the set of possible outcomes.

Answer (Please select your correct option)

VuAnswers.com

☐ event

☐ sample space

☐ subset

correct

Made by: Waqar Siddhu

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

VuAnswers.com

Answer (Please select your correct option)

☐ Dependent

correct

☐ Independent

☐ Exclusive

Made by: Waqar Siddhu

A rule that assigns a numerical value to each outcome in a sample space is called

VuAnswers.com

Answer (Please select your correct option)

☐ One to one function

☐ Conditional probability

☐ Random variable

correct

Made by: Waqar Siddhu

The expectation of x is equal to

VuAnswers.com

Answer (Please select your correct option)

Sum of all terms



Sum of all terms divided by number of terms



correct

$\sum xf(x)$



Made by: Waqar Siddhu

The expectation μ for the following table is

x_i	1	3
$f(x_i)$	0.4	0.1

Answer (Please select your correct option)

VuAnswers.com

☐

0.5

☐

3.4

☐

0.3

☐

0.7

Made by: Waqar Siddhu

Two distinct edges with the same set of end points are called

VuAnswers.com

Answer (Please select your correct option)

Isolated

☐

Incident

☐

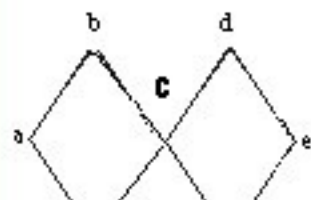
Parallel

☐

correct

Made by: Waqar Siddhu

The graph given below



Answer (Please select your correct option)

VuAnswers.com

has not Euler circuit

☐

has Hamiltonian circuit

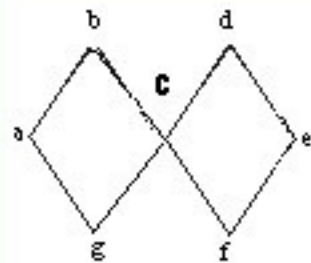
☐

does not have Hamiltonian circuit

☐

correct

Made by: Waqar Siddhu

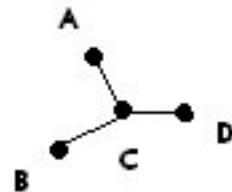


Answer (Please select your correct option)

VuAnswers.com

☐ has not Euler circuit☐ has Hamiltonian circuit☐ does not have Hamiltonian circuitcorrect**Made by: Waqar Siddhu**

In the given graph, vertex "C" has degree



Answer (Please select your correct option)

VuAnswers.com

☐ 2

☐ 3

correct

☐ 4

☐ 6

Made by: Waqar Siddhu

Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does the graph have?

Answer (Please select your correct option)

VuAnswers.com

☐ 12

correct

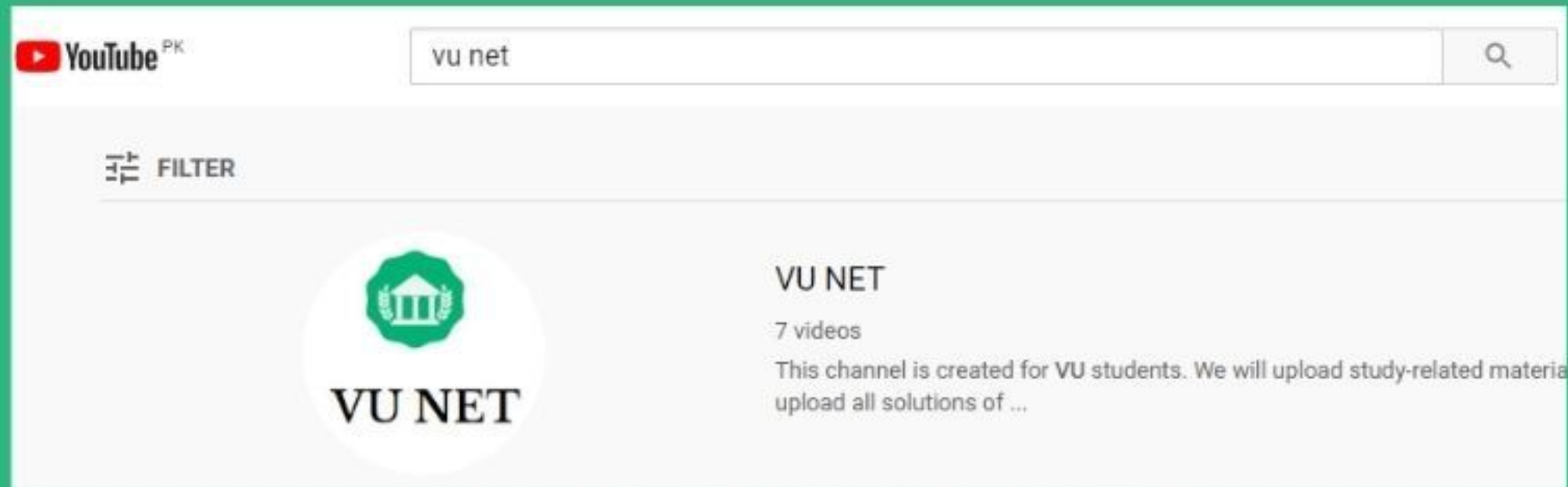
☐ 13

☐ 14

☐ 8

Made by: Waqar Siddhu

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